DATA BRİCKS ASSET BUNDLES

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About DABs

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- Databricks Asset Bundles (DABs) are a way to manage Databricks resources using code.
- They allow you to define jobs, workflows, and pipelines in YAML format.
- Enable automated deployment using Databricks CLI.
- Support version control with Git.
- Used for production-grade data engineering projects.

Importance

Without DABs:

- Manual configuration in UI
- No proper version tracking
- Difficult Dev → Test → Prod movement
- Higher risk of production errors

With DABs:

- Structured deployment
- Controlled environment management
- Safe and repeatable releases

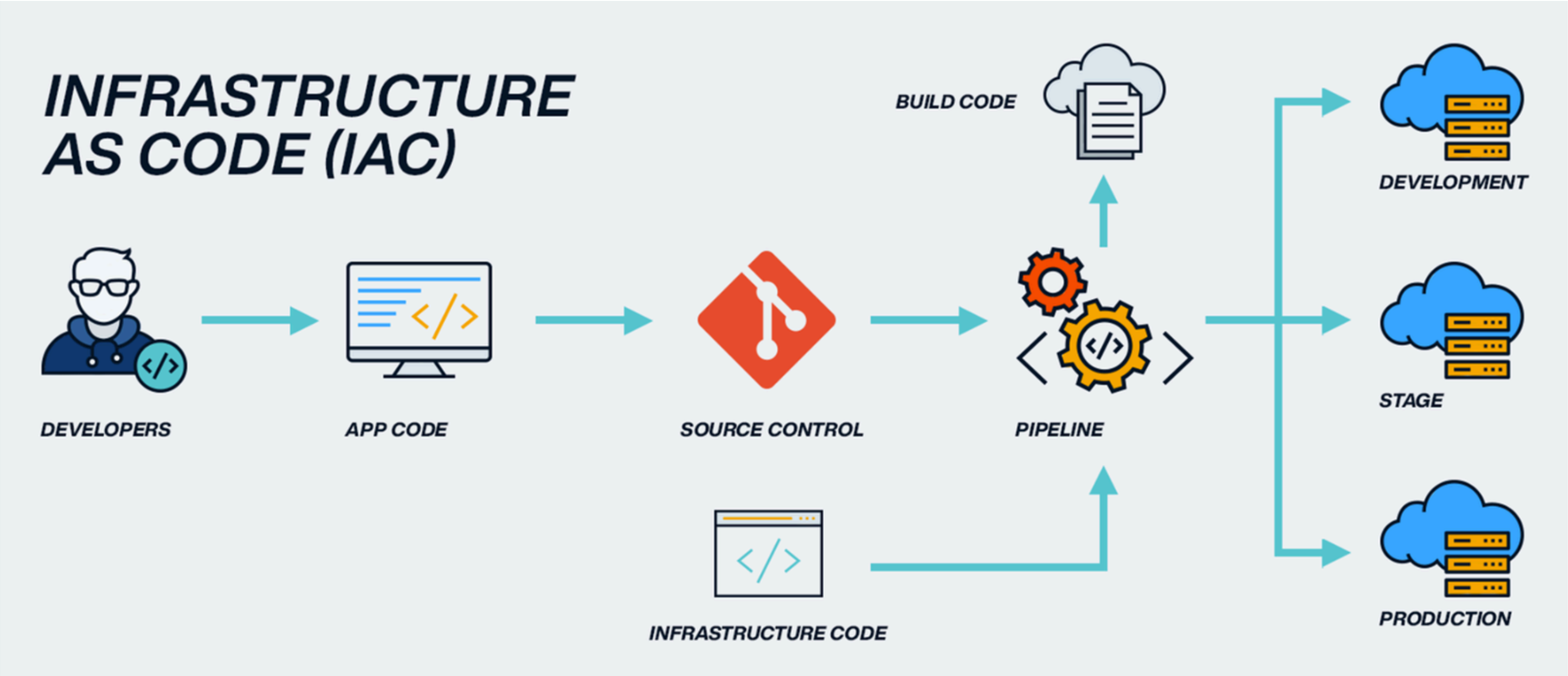
Structure

A typical DAB project contains:

- databricks.yml → Main configuration file
- resources/ → Job & pipeline definitions
- src/ → Notebooks / Python scripts
- targets → Dev, Test, Prod environments

```
my-dab-project/
│
├── databricks.yml
├── resources/
│     ├── job.yml
│     └── pipeline.yml
│
├── src/
│     └── main.py
│
└── tests/
```

DAB Workflow



Creation And Deployment

```
bundle init
      ↓
bundle validate
      ↓
bundle deploy
      ↓
bundle run
```

Using Databricks CLI:

- bundle init → Create bundle structure
- bundle validate → Check configuration
- bundle deploy → Deploy to workspace
- bundle run → Execute job

Thank you